Personal information

Romain Claret
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Research Interests

Neuroevolution · Artificial Life · Bio-Inspired · Open-Endedness · Behaviors · Space · Robotics

Education

2020 – 2026 — **Ph.D. Computer Science** with Prof. Dr. Kilian Stoffel and Prof. Dr. Paul Cotofrei.
Scaling Adaptive Substrate Neuroevolution.
University of Neuchâtel, Neuchâtel, Switzerland

2018 – 2020 — **M.Sc. HES-SO Software Engineering** with Prof. Dr. Jean Hennebert.
Multi-hop Multi-turns Question-Answering Chatbot using Sub-Knowledge Graphs.
University of Applied Sciences and Arts of Western Switzerland, Lausanne, Switzerland

2013 – 2016 — **B.Sc. HES-SO Software Engineering** with ing. info. dipl. EPF Marc Schaefer.
Anonymous and Decentralized Browser-based Data Sharing Service.
University of Applied Sciences and Arts of Western Switzerland, Neuchâtel, Switzerland

Grants & Fellowships

2023 – 2024 — **Doc.Mobility Fellowship** *(CHF 22'778)*. Awarded by swissuniversities and the University of Neuchâtel to fund a 6-month research stay (Sep 2023 – Feb 2024) at the Natural Computing Research & Applications group (NCRA).
University College Dublin, Dublin, Ireland.

Complementary Education

2024 – 2025 — **GECCO Summer School** *Computer Science Workshops* (6 days).
RMIT University, Melbourne, Australia / University of Málaga, Málaga, Spain

2022 – 2024 — **BENEFRI Summer School** *Computer Science Workshops / Presenting* (6 days).
University of Neuchâtel, Bern, and Fribourg

2023 — **CUSO Winter School** *Computer Science Workshops / Presenting* (28 Hours).
University of Neuchâtel, Switzerland

2020 – 2024 — **CUSO Doctoral Training** *Computer Science and Skills* (112 Hours).
University of Neuchâtel, University of Geneva, University of Lausanne, Online

2020 — **Required Doctoral Coursework** *MSc level Computer Science classes* (18 ECTS).
Machine Learning, Computational Thinking, Programming, Business Analytics.
University of Neuchâtel (UNINE), Neuchâtel, Switzerland

2010 – 2012 — **Undergraduate** *Major in Physics* (5 ECTS).
Swiss Federal Institute of Technology Lausanne (EPFL), Lausanne, Switzerland

2009 – 2010 — **Undergraduate** *Major in Physics & Mathematics* (35 Undergraduate Credits).
Northeastern University, Boston, Massachusetts, United States of America

2009 — **Extension School** (4 Undergraduate Credits).
Followed the class of Prof. Dr. Wolfgang Rueckner, Principles of Physics IT (PHYS E-1B).
Harvard University, Cambridge, Boston, Massachusetts, United States of America

2008 — **Summer School** (8 Undergraduate Credits).
Followed the class of Dr. Jonathan Weintroub, Laboratory Electronics (PHYS S-123).
Harvard University, Cambridge, Boston, Massachusetts, United States of America

Experience

2026 – ····
University Teaching Specialist at the UCD Smurfit School of Business. Conducting research at the UCD Natural Computing Research & Applications Group with Prof. Michael O'Neill alongside teaching duties.
University College Dublin, Dublin, Ireland.

2023 – 2025
Visiting Researcher at UCD Natural Computing Research & Applications Group. (Supervised by Prof. Michael O'Neill).
University College Dublin, Dublin, Ireland.

2020 – 2026
Doctoral Assistant at the Information Management Institute. Applied Mathematics & Databases for 1st-year Bachelor. (Supervised by Dr. Paul Cotofrei).
University of Neuchâtel, Neuchâtel, Switzerland.

2017 – 2018
IT Independent. Consulting in Software Engineering and Blockchain. The tasks were to advise technologies, define projects, make prototypes, and conduct workshops.
Geneva area, Lausanne area, and Versicherix AG, Switzerland.

2016 – 2017
Co-Founder at a Blockchain-based InsurTech Startup. Lead in blockchain and innovation. The tasks were to make software architectures, high-level product schematics, management, workshops, prototypes, and documentation. Administration, fundraising, exhibitions, partnerships, and market studies.
Versicherix AG, Solothurn, Switzerland.

2010 – 2015
Founder at a Multimedia Streaming Startup. Lead developer and executive. The tasks were to make software architectures, implement prototypes & software, manage digital rights, build the business model, perform market studies, conduct fundraising, create partnerships with film studios, and comply with copyright laws.
Libacy, Neuchâtel, Switzerland.

2012
Summer Internship in Watchmaking R&D using 3D computer-aided design.
Manufacture Claret, Le Locle, Switzerland

2010
Summer Internship at Jenks Vestibular Physiology Lab under the supervision of Asst. Prof. Dr. Faisal Karmali. Design an experiment to identify a link between vision and the vestibular system. The tasks were to build the setup in 3D, build the setup by adapting a hydraulic flight simulator, run Matlab simulations, experiment on human subjects, and interpret the results.
Massachusetts Eye & Ear Infirmary, Harvard Medical School, Boston, United States of America

2004 – 2006
Summer Internships in Watchmaking: 3D computer-aided design (CAD) construction and technical drawing of watch movements (2004), disassembly, reassembly and customization of a mechanical pocket watch (2005), manufacturing, chamfering, and technical control (2006).
Manufacture Claret, Le Locle, Switzerland

Teaching

2026 – ····
University Teaching Specialist. Programming for Analytics.
Trimester 1 class given to master students. (approx. 150 students).
University College Dublin, Dublin, Ireland

2025
Computer Science Master's Thesis Supervisor. Evolutionary Computation Hyperparameter Optimization for Neuroevolution.
University of Neuchâtel, Neuchâtel, Switzerland

2020 – 2026
Teaching Assistant. Applied Mathematics (Analysis and Linear Algebra).
2 hours lecture and 2 hours QA per week per semester given to first-year bachelors in Economical Science and Data Science. (approx. 150 to 200 students per semester).
University of Neuchâtel, Neuchâtel, Switzerland

Teaching Assistant. Databases (Modelization, SQL, NoSQL, Visualization).
2 hours QA and student's project management per week during spring semester given to first-year and third-year bachelors in Data Science and Economical Science. (approx. 60 students per semester).
University of Neuchâtel, Neuchâtel, Switzerland

Teaching (continued)

2020 – 2022 **Guest Lecturer.** "Demystifying Artificial Intelligence for Health Professionals".
3 hours yearly lecture given to second-year physicians. (approx. 20 students).
University of Geneva, Geneva, Switzerland

2021 **Teaching Assistant.** Digital Management and Processing of Multimedia Data (Analyze data, Identify and Formulate issues, Create visualizations).
4 days of project supervision and QA during the fall semester (block course) given to bachelors in Economical Science and Law. (approx. 40 students).
University of Neuchâtel, Neuchâtel, Switzerland

Activities in panels and commissions

2025 – 2026 **UniNE**. Member of the University Assembly of the University of Neuchâtel.

2023 – 2026 **UniNE**. Committee Member of the Mid-level staff Association (ACINE).

2015 – 2017 **HES-SO**. Member of the Participatory Council of the Engineering domain.

Memberships

Scientific & Industry Societies

2025 – .... **SRA**. Member of the Swiss Robotics Association.

2022 – .... **IEEE**. Member of the Institute of Electrical and Electronics Engineers.

ACM. Member of the Association for Computing Machinery.

ISAL. Member of the International Society for Artificial Life.

Student Societies

2023 – 2026 **UniNE**. President of the Computational Thinking Association (Algorithmia).

UniNE. Treasurer of the Association Suisse Alémanique & Friends (ASA&F).

2014 – 2016 **HES-SO**. Head of Communication of the Student Association (REH-SO).

2010 – 2012 **EPFL**. Webmaster of the Role-playing Association (JDRpoly).

EPFL. Committee Member of the Physics Students Association. (Irrotationnels)

EPFL. Member of the Robotics Association (RoboPoly).

2009 – 2010 **Northeastern University**. Member of the Society of Physics Students (SPS).

Skills

Languages	French Native, English Fluent, Russian Proficient, German Basic
Code & Frameworks	Python, JAX, Pytorch, NetworkX, Scikit, C++, JS, ...
Machine Learning	Evolutionary, Deep learning, Edge Models, LLMs API & Local, ...
Computing	Optimization, HPC, Graphs, Kubernetes, Distributed, Parallel, ...
Dev-Ops	Git, Docker, Continuous Integration, IaaS, PaaS, SaaS, ...
Miscellaneous	CAD, Electronics, 3D Printing, Knowledge Bases, LaTeX, Blockchain, ...

Certifications

2013 **Entrepreneurship**. Granted by Venturelab, Switzerland.

2012 **Inventor Training** (CAD). Granted by Hurni Engineering, Switzerland.

2008 **Human Subjects Research**. Granted by CITI program, USA.

Publications

Peer-Reviewed Publications

[1] **R. Claret**, M. O'Neill, P. Cotofrei, and K. Stoffel, "Tensor-Accelerated Eager Multi-Resolution Grids for Evolving Large-Scale Substrates," in *Proceedings of the Genetic and Evolutionary Computation Conference Companion*, San Jose, Costa Rica: ACM, 2026. DOI: `10.1145/3795101.3805361`. Online PDF.

[2] **R. Claret**, A. Gygax, M. O'Neill, P. Cotofrei, and P. Felber, "Early-Stopping Thresholds for ES-HyperNEAT: A Data-Driven Approach from Fitness Dynamics," in *Proceedings of the IEEE Congress on Evolutionary Computation (CEC)*, To appear, IEEE, 2026. Online PDF.

[3] **R. Claret**, M. O'Neill, P. Cotofrei, and K. Stoffel, "Investigating Hyperparameter Optimization and Transferability for ES-HyperNEAT: A TPE Approach," in *Proceedings of the Genetic and Evolutionary Computation Conference Companion*, Melbourne, VIC, Australia: ACM, 2024, pp. 1879–1887. DOI: `10.1145/3638530.3664144`. Online PDF.

[4] **R. Claret**, A. Gygax, M. O'Neill, P. Cotofrei, M. Palma Mendes, and P. Felber, "Breaking the Central Bias: Spatially Partitioned Experts for ES-HyperNEAT," To appear, 2026. Online PDF.

[5] **R. Claret**, M. O'Neill, P. Cotofrei, and K. Stoffel, "Multi-Behavioral Evolved Substrates Through Neuromodulation and Activation Selection," To appear, 2026. Online PDF.

[6] **R. Claret**, M. O'Neill, P. Cotofrei, and K. Stoffel, "Per-Node Activation Function Evolution in Indirectly Encoded Substrates: Solvability, Limits, and Emergent Diversity," To appear, 2026. Online PDF.

[7] **R. Claret**, M. O'Neill, P. Cotofrei, and K. Stoffel, "Bio-Inspired Meta-Learning Strategies for Per-Node Activation Function Discovery in Evolved Neural Substrates," To appear, 2026. Online PDF.

Manuscripts Under Review

[8] **R. Claret**, M. O'Neill, P. Cotofrei, and K. Stoffel, "Persistent Populations Enable Continual Learning in Evolved Neural Substrates," Under review, 2026.

[9] **R. Claret**, M. O'Neill, P. Cotofrei, and K. Stoffel, "Per-Node Aggregation Function Evolution in Indirectly Encoded Substrates: Asymmetric Complementarity with Activation," Under review, 2026.

[10] **R. Claret**, M. O'Neill, P. Cotofrei, and K. Stoffel, "Co-Evolving Neurotransmitter Vectors in Indirectly Encoded Substrates: A Search Landscape Characterization," Under review, 2026.

Preprints, Abstracts & Posters

[11] **R. Claret**, M. O'Neill, P. Cotofrei, and K. Stoffel, "The Quadtree Bottleneck in ES-HyperNEAT: From GPU Barriers to Alternative Discovery Strategies," Available as a preprint on arXiv, 2026. Online PDF.

[12] **R. Claret**, *Scaling Neuroevolution for Heterogeneous Tasks using Universal Knowledge Representation*, Poster and ongoing-research presentation, ACM International Conference on Distributed and Event-Based Systems (DEBS), Neuchâtel, Switzerland, 2023. Online PDF.

[13] F. Karmali, K. Lim, A. Adatia, **R. Claret**, K. Nicoucar, and D. M. Merfeld, *Perceptual Roll-Tilt Thresholds Demonstrate Visual-Vestibular Fusion*, Conference abstract and poster, 40th Annual Meeting of the Society for Neuroscience, San Diego, CA, USA, Nov. 2010. Online PDF.